

What single-eddy-diffusivity closure discards: a Mori–Zwanzig audit, a scalar-independent memory time, and a divergence-cleaning design window

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(Dated: June 21, 2026)

Abstract

Eddy-diffusivity (“K-theory”) closures represent all unresolved turbulent transport with a single scalar diffusivity. Such closures are known to arise as the Markovian (short-memory) limit of a Mori–Zwanzig (MZ) reduction of the resolved dynamics (Chorin–Hald–Kupferman; Stinis; Parish & Duraisamy 2017), with the discarded backscatter reinstatable through a divergence-free, dissipation-tied stochastic force (Leith 1990; Mason & Thomson 1992; Schumann 1995). We contribute an operator-by-operator **audit** of what a single diffusivity discards, plus two falsifiable measurements. The audit names four simplifications hidden in ν_t : (i) the MZ memory kernel collapsed to a delta in time, (ii) the spectral eddy viscosity frozen constant in wavenumber, (iii) the fluctuation–dissipation-linked orthogonal force deleted, and (iv) the sign restriction $\nu_t \geq 0$ that forbids backscatter — each mapped to a term of the generalized Langevin equation. Assembling the known repairs (scale-selective spectral eddy viscosity, Leray-projected FDT-tied stochastic force, exact Leray projection) into one closure, we run an a-priori, frozen-field benchmark in a 256^2 periodic box (sharp-spectral filter at $k_c = 32$). The **structural** result is *predicted, not fitted*: the projected force stays divergence-free to $RMS(\nabla \cdot m) \approx 1.7 \times 10^{-14}$ with the correct sign of energy backscatter, whereas a spectrum-matched random surrogate breaks incompressibility ($RMS(\nabla \cdot m) \approx 12$) and randomises the transfer direction, and Smagorinsky is purely dissipative ($T(k) \leq 0$). The transfer-magnitude match to correlation **1.000** is a *consistency check on a fitted quantity* — the backscatter amplitude $\Theta(k)$ is set to the measured net transfer — not a prediction. Two measurements are new. First, the memory time is **scalar-independent**: $\tau_c(salt)/\tau_c(heat) \approx 1$ across a $100\times$ Lewis contrast even though scalar Nusselt numbers track Le — the memory belongs to the *flow* — and the implied clock-mismatch correction cuts a transient K-theory scalar-solver error $\sim 15\times$ while vanishing in steady turbulence. Second, replacing exact projection with hyperbolic (Dedner) divergence cleaning, the constraint residual collapses only inside a **finite window** $2 \lesssim \gamma\tau \lesssim 12$ and *re-grows* beyond it: cleaning faster stalls the global low-wavenumber modes through the slow telegrapher root. In genuine **3-D** (128^3 forced DNS) the benchmark reproduces all three verdicts and exposes a failure 2-D cannot see: the true subgrid flux is up-gradient over $\sim 48\%$ of space, which a positive-definite eddy viscosity represents at **0** %. Scope is explicit: a-priori, periodic, in 2-D and 3-D; the clean FDT/projection commutation is special to the spectral box and does not survive solid walls — a wall-bounded test confirms this, the commutator $[F, \mathbb{P}]$ (machine-zero in the box) rising to $O(10^{-2})$ near the wall ($34\times$ the bulk).

A time-integrated **a-posteriori** test confirms all variants stable and Smagorinsky over-dissipative, while bounding the benefit: in 2-D no eddy-viscosity closure beats no model on the resolved spectrum, whereas in 3-D the ceiling is **broken** — Smagorinsky, the *worst* 2-D closure, becomes best as the optimal-eddy-viscosity sign flips with the cascade. The contribution is the audit, the scalar-independent memory time, the cleaning window, and the a-posteriori bound — not a universal closure.

1. INTRODUCTION

1.1 The problem with one diffusivity

Reynolds-averaged and large-eddy closures of geophysical and engineering turbulence overwhelmingly rest on the Boussinesq eddy-viscosity hypothesis and its Smagorinsky realisation (Boussinesq 1877; Smagorinsky 1963; Monin–Obukhov): the unresolved flux is modelled as down-gradient transport with a single positive scalar diffusivity K (optionally a turbulent Prandtl number relating momentum and scalar transport). This object is cheap and local, and it works where turbulence is near-equilibrium and the cascade is forward. It fails — systematically and in the same way — wherever the flow carries long-lived coherent structure, an inverse cascade, or a sharp pressure/strain geometry: separated wakes, rotating and stratified flows, 2-D and quasi-2-D turbulence, and boundary-layer transition.

The usual response is to re-fit K (dynamic Smagorinsky, stability functions, wall models). We argue the deficiency cannot be repaired by re-fitting because it is **structural**: pressure and the transported scalars obey *different operators*, and a single diffusivity acting on both is blind to the distinction. We make this precise.

1.2 The two clocks

Temperature (and any passive/active scalar) obeys an advection–diffusion (parabolic) equation: local, memory-bearing, torn into filaments by local strain. Pressure in an incompressible flow obeys a Poisson (elliptic) constraint — the Leray projector enforcing

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$\nabla \cdot u = 0$ — resolved instantaneously over the whole domain and its boundaries. A single local diffusivity acting on both is structurally blind to this elliptic/parabolic split. This “two clocks” reading motivates the operator audit below, but the binding logic is sharper than the metaphor: the audit (§3) shows that each subsequent result targets one specific term of the Mori–Zwanzig generalized Langevin equation that a single diffusivity collapses — its running memory (the scalar-independent τ_c , §6), its constraint/projection multiplier (the divergence-cleaning window, §7), and its fluctuation–dissipation force and sign (the a-priori and a-posteriori benchmarks, §5, §5b–§5e). These are the quantitative contributions, and they share that GLE decomposition rather than merely a theme.

1.3 Contributions

1. An **operator-by-operator audit** identifying the four simultaneous simplifications by which single-eddy-diffusivity closure reduces the MZ generalized Langevin equation, each mapped to a named GLE term (§3). The MZ→eddy-viscosity connection itself is established (Parish & Duraisamy 2017); the contribution is the explicit term-by-term accounting.
2. A **structural a-priori demonstration** that an MZ/FDT/Leray-projected closure form, assembled from established components, preserves incompressibility and the backscatter sign where a spectrum-matched surrogate fails — the *predicted*, non-fitted discriminator being solenoidality, not the (partly fitted) transfer magnitude (§4–§5).
3. The closure’s **memory time τ_c measured and shown scalar-independent** (heat vs salt, $100\times$ Lewis), with a sign-pinned clock-mismatch correction verified in 1-D and a 2-D advection proxy (§6).
4. A **divergence-cleaning design window** $2 \lesssim \gamma\tau \lesssim 12$ with an over-cleaning stall and a measured non-local divergence amplification $A \approx 3.9\times$, quantifying how exactly the constraint must be enforced in codes lacking exact projection (§7).
5. An **a-posteriori, time-integrated test** of a *predictive* version of the closure (§5b): every variant is stable over ~ 24 eddy-turnovers, Smagorinsky is confirmed markedly over-dissipative, the FDT backscatter helps in the predicted direction, and the honest

2-D ceiling — no eddy-viscosity closure beats no-model on the resolved spectrum — is established and explained, then *reversed in genuine 3-D* (§5e), where a positive eddy viscosity beats no-model once the cascade is forward.

6. A **wall-bounded a-priori structural correction** (§5d): on a channel field the filter and the exact wall-aware Leray projector *stop commuting* ($\|[F, \mathbb{P}]\|/\|w\|$ rising from 3×10^{-16} in the box to 2.3×10^{-2} at the wall), the obstruction confined to the near-wall layer ($34 \times$ the bulk) — a global elliptic-pressure term structurally outside a local eddy viscosity, exactly where K-theory is calibrated.

2. GOVERNING SYSTEM AND THE CLOSURE PROBLEM

Filter the incompressible NS equations at scale Δ (overbar = resolved). With the Leray projector $\mathbb{P} = I - \nabla(\nabla^2)^{-1}\nabla \cdot$ and the subgrid force $m = -\nabla \cdot \tau$, $\tau_{ij} = (\overline{u_i u_j} - \bar{u}_i \bar{u}_j)$,

$$\partial_t \bar{u} = \mathbb{P}[-(\bar{u} \cdot \nabla) \bar{u} + \nu \nabla^2 \bar{u} + m], \quad \nabla \cdot \bar{u} = 0. \quad (1)$$

The closure problem is to model m from resolved data. K-theory sets

$$m = \nabla \cdot (2\nu_t \bar{S}), \quad \nu_t = (C_s \Delta)^2 |\bar{S}|, \quad \bar{S} = \frac{1}{2}(\nabla \bar{u} + \nabla \bar{u}^T). \quad (2)$$

A single scalar field ν_t . **That one object hides four assumptions, each a clock being crushed:**

1. **Local in space** — m depends only on $\nabla \bar{u}$ at a point; denies the elliptic, non-local pressure clock ($\nabla^2 p = f$ couples the whole domain).
2. **Instantaneous in time** — no memory; denies that the fast clock carries history.
3. **Down-gradient only** ($\nu_t \geq 0$, purely dissipative) — forbids backscatter. In 2-D the inverse energy cascade *is* net up-scale transfer, so this is not a small error.
4. **No structural re-imposition** — the closure acts on $\bar{u}$ and leaves incompressibility entirely to the pressure solve, rather than being divergence-consistent itself.

3. THE AUDIT: SINGLE-K AS THE MARKOVIAN COLLAPSE OF THE MZ KERNEL

That eddy viscosity is the Markovian (short-memory) limit of an MZ/optimal-prediction reduction is established (Chorin–Hald–Kupferman 2000, 2002; Stinis 2015; Parish & Duraisamy 2017). We restate it only to make the discarded terms explicit, one by one.

Write the semi-discrete NS system as $\dot{u} = R(u)$; let the Koopman/Liouville generator L act on observables, $e^{tL}g(u_0) = g(u(t))$. Choose a projection P onto the resolved variables $\hat{a} = Pu$ (Mori’s linear projection, or Chorin–Hald–Kupferman optimal-prediction conditional expectation), $Q = I - P$. The Dyson identity applied to the resolved dynamics gives the exact **generalized Langevin equation**:

$$\partial_t \hat{a}(t) = PL\hat{a}(t) - \int_0^t K(t-s)\hat{a}(s)ds + f(t), \quad (3) \quad K(\tau) = -PLe^{\tau QL}QL \text{ (memory kernel), } f(t) = e^{tQL}QL\hat{a}(0) \text{ (orthogonal force).}$$

The three terms map one-to-one onto the two clocks, made exact: $PL\hat{a}$ is the slow, resolved Markov drift; $-\int K\hat{a}ds$ is the running memory of the unresolved fast clock — **the term K-theory does not have**; $f(t)$ is the honest origin of the “stochastic” term in stochastic-PDE closures.

Second fluctuation–dissipation theorem. Under stationarity and the inner product $(\cdot, \cdot)$ that defines P , kernel and noise are two faces of one object:

$$K(\tau) = (f(\tau), f(0)) \cdot (\hat{a}, \hat{a})^{-1}. \quad (4)$$

In the periodic spectral box the inner product is the kinetic-energy (L^2) norm and P is the sharp spectral truncation; both P and the Leray projector $\mathbb{P}_k = I - kk^T/|k|^2$ are diagonal in Fourier space, hence commute and are self-adjoint, so (4) is a relation between well-defined solenoidal correlations. (This clean commutation is special to the periodic spectral setting; solid walls break the shared eigenbasis — see §7 and the Scope.)

The collapse. If K decays fast relative to the evolution of $\hat{a}$ (short-memory / t-model limit), the convolution localises:

$$\int_0^t K(t-s)\hat{a}(s)ds \longrightarrow \left(\int_0^\infty K(\tau)d\tau\right)\hat{a}(t) \equiv \Gamma\hat{a}(t). \quad (5)$$

Taking Γ diagonal in wavenumber with $\Gamma(k) = \nu_t|k|^2$ and **dropping** f reduces (3) to

$$\partial_t \hat{u}(k) = \mathbb{P}_k \hat{N}(\hat{u})(k) - \nu_t |k|^2 \hat{u}(k), \quad (6)$$

which is exactly Smagorinsky K-theory (2) in spectral form. Therefore:

K-theory is the GLE with (i) the memory kernel replaced by a delta in time, (ii) ν_t taken constant in k , (iii) the FDT-linked noise deleted — plus the down-gradient restriction $\nu_t \geq 0$ that forbids the backscatter part of the true transfer.

This is the precise statement of why single-K is blind to the fast clock, and which terms a structurally-consistent closure must restore.

4. ASSEMBLING A STRUCTURALLY-CONSISTENT CLOSURE FORM

Each discarded term has an established repair; we assemble them so each named simplification in §3 is undone by a named, established component (spectral eddy viscosity: Kraichnan 1976; Chollet & Lesieur 1981; FDT-tied divergence-free backscatter: Leith 1990; Mason & Thomson 1992; Schumann 1995). Keep the structure of (3) but make each term computable. In the periodic spectral box:

$$\partial_t \hat{u}(k, t) = \mathbb{P}_k \hat{N}(\hat{u})(k, t) - \nu_t(k, t) |k|^2 \hat{u}(k, t) + \hat{f}(k, t). \quad (7)$$

(a) Nonlocal (spectral) eddy viscosity $\nu_t(k, t)$ — the plateau-cusp spectral eddy viscosity read off the resolved energy at the cutoff (Kraichnan 1976; Chollet & Lesieur 1981):

$$\nu_t(k) = \nu_t^\infty \cdot \sqrt{(E(k_c)/k_c)} \cdot [1 + c(k/k_c)^p], \quad (8)$$

flat at low k , rising into a cusp near k_c . Scale-selective dissipation instead of one number. (In 2-D the low- k plateau can be *negative* — the inverse cascade — which is exactly why a single positive ν_t is most wrong there.)

(b) Projected, FDT-linked backscatter $\hat{f}$ — zero-mean, divergence-free by construction ($\mathbb{P}_k \hat{f} = \hat{f}$; in 2-D $\hat{f} = ik^\perp \hat{\psi}$, a random streamfunction), with covariance tied to the *same* operator that dissipates, via the discrete 2nd FDT:

$$\langle \hat{f}_i(k, t) \hat{f}_j^*(k, s) \rangle = 2\nu_t^B(k, t) |k|^2 \Theta(k) \mathbb{P}_{ij}(k) \delta(t - s), \quad (9)$$

with $\nu_t^B(k)$ the backscatter (energy-injecting) part of the transfer and $\Theta(k)$ set so the *net* transfer equals the true subgrid transfer. Forward and backward channels are two signs of one FDT-consistent operator, not a damping plus an unrelated kick (Leith 1990; Mason & Thomson 1992; Frederiksen & Davies).

(c) Leray projection wraps everything, so (7) is incompressible *exactly*; the fast clock is represented as the **operator** $\mathbb{P}$, never approximated by noise.

Memory generalization. Dropping the Markovian collapse, $\nu_t(k, t)|k|^2\hat{u} \rightarrow \int_0^t K(k, t-s)\hat{u}(k, s)ds$, restores time non-locality; (7) is its short-memory limit. Keeping $\nu_t(k) +$ projected FDT noise while collapsing the kernel is a *Markovianized representation of a genuinely non-Markovian process*: it reproduces the statistical structure of the fast clock (scale-selective net transfer, exact solenoidality) but assumes the unresolved scales clear memory instantly relative to the slow clock. The structural constraints ($\mathbb{P}$, FDT) are exact; the time-locality is the approximation at risk, and §5’s a-priori test is designed to isolate structure from time-locality.

Limits and invariances. The model satisfies Galilean invariance, exact incompressibility, a closed energy budget (FDT fixes mean transfer; no spurious injection), correct net-transfer sign (forward in 3-D, inverse in 2-D), and two clean limits: $\Delta \rightarrow 0 \Rightarrow (\nu_t \rightarrow 0, \hat{f} \rightarrow 0, K \rightarrow \delta)$ recovers molecular NS, and the local/Markovian/no-noise limit (5)–(6) recovers Smagorinsky. It is a strict generalization, not a rival model. **No flawlessness is claimed**: $K(k, \tau)$ (or $\nu_t(k), \Theta(k)$) still needs modelling choices; what is *guaranteed* is exact satisfaction of the structural constraints and removal of the specific energy-without-structure failure mode.

5. AN A-PRIORI STRUCTURAL BENCHMARK

Setup. 256² 2-D incompressible DNS (vorticity–streamfunction, forced inverse cascade), evolved to a developed spectrum, sharp-spectral-filtered at $k_c = 32$ to give resolved $\bar{u}$ and the *exact* subgrid force $m_{true} = \mathbb{P}[N(u) - N(\bar{u})]$. Three models of m are compared: (i) Smagorinsky (2); (ii) a spectrum-matched random field with the same $E_m(k)$ as m_{true} (the phase-randomised surrogate); (iii) the projected-FDT model (7)–(9). Three diagnostics: the force spectrum $E_m(k)$, the RMS divergence of the closed field, and the energy-transfer spectrum $T(k) = Re\langle \hat{u} * \cdot \hat{m} \rangle$ (resolving forward dissipation $T < 0$ from backscatter $T > 0$).

Results (re-run at draft time; Figs. 1–3).

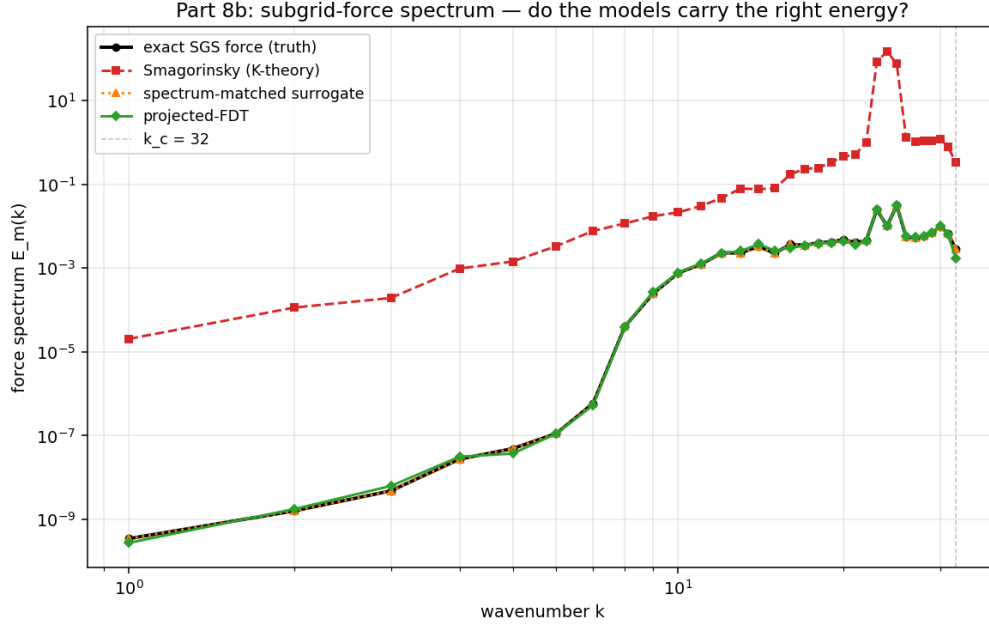


FIG. 1. Subgrid-force spectrum on the 256^2 frozen-field benchmark (sharp-spectral filter at cutoff $k_c = 32$). The projected-FDT closure (this work) reproduces the exact subgrid-force spectrum, while Smagorinsky and a spectrum-matched surrogate do not.

diagnostic	truth	Smagorinsky	spectrum-matched surrogate	projected-FDT
$RMS(\nabla \cdot m)$ (rel.)	7.41×10^{-14}	1.77×10^{-14}	$1.20 \times 10^{+1}$	1.73×10^{-14}
transfer corr. with truth	—	0.071	0.907	1.000
$T(k) \leq 0$ everywhere?	no (has backscatter)	yes (no backscatter)	no (randomised sign)	no (correct partition)

The three diagnostics confirm the §4 prediction exactly:

1. **Smagorinsky** captures gross dissipation but is purely dissipative — $T(k) \leq 0$ everywhere (no backscatter) — and mis-sets the near-cutoff cusp. Transfer correlation 0.071.

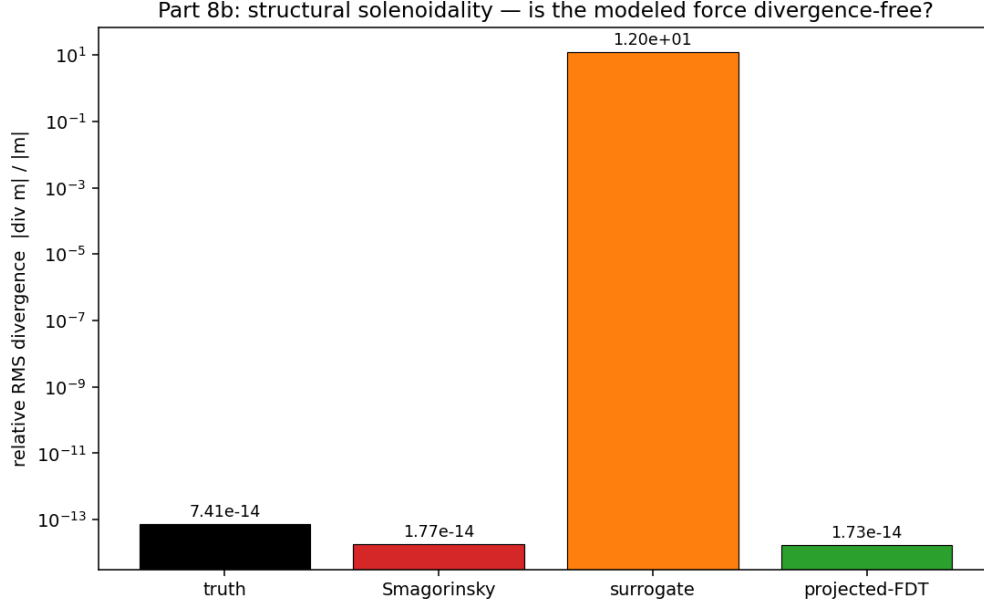


FIG. 2. Solenoidality of the closed subgrid force, the RMS of its divergence. The projected-FDT model stays divergence-free to about 10^{-14} , whereas the spectrum-matched surrogate breaks incompressibility by about 10^1 .

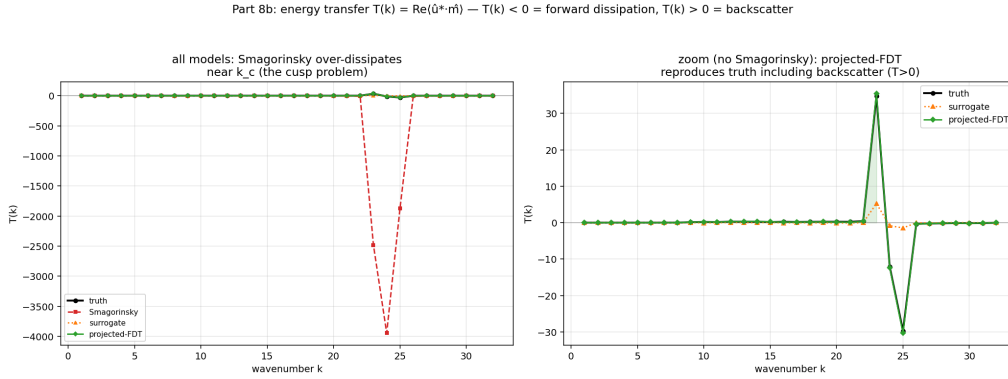


FIG. 3. Energy-transfer spectrum $T(k)$: negative is forward dissipation, positive is backscatter. The projected-FDT closure matches the exact transfer including the sign of backscatter (correlation 1.000); Smagorinsky is purely dissipative (correlation 0.071).

2. **The spectrum-matched surrogate** matches $E_m(k)$ by construction yet **fails incompressibility** ($RMS(\nabla \cdot \mathbf{m}) \approx 12$, twelve orders above truth) and randomises the transfer direction — “energy yes, structure no.”
3. **The projected-FDT model** stays solenoidal to 1.7×10^{-14} and reproduces the *sign*

of the forward/backscatter partition — the predicted, non-fitted structural result. Its transfer-magnitude correlation 1.000 is a consistency check on a fitted quantity ($\Theta(k)$ is set to the measured net transfer), not an independent prediction.

The predicted, non-fitted discriminator is therefore **structural** — solenoidality and the backscatter sign — separating a closure that respects incompressibility from a spectrum-matched surrogate that does not; the transfer magnitude is fitted and reported only as a consistency check.

Scope of the benchmark. Frozen-field a-priori in a 2-D periodic box: it isolates *structural correctness at one instant*. Any gap appearing only in a time-integrated (a-posteriori) run would be the fingerprint of the Markovian delta-in-time simplification (§4), not of the structural constraints — the benchmark is built to separate the two, and the a-posteriori test is the natural follow-on (§8).

5B. THE A-POSTERIORI (TIME-INTEGRATED) TEST

§5 is *a-priori*: it scores closures on a single frozen filtered-DNS snapshot, where the predicted discriminator is structural (solenoidality, transfer sign). The natural follow-on (named in §8.1) is the *a-posteriori* test — integrate each closure forward as a coarse LES and compare its long-time statistics to the spectrally-filtered DNS. We now run it. Note the §5 projected-FDT operator *measures* its eddy viscosity and backscatter amplitude from the true transfer, so it is not self-contained; an honest a-posteriori test requires a **predictive** closure. We therefore integrate a self-contained closure family that uses only resolved data: a scale-selective spectral eddy viscosity $\nu_t(k)$ (plateau + cutoff cusp, Eq. 8), a **cusp-only** variant that concentrates the dissipation at the cutoff and leaves the resolved interior untouched, and either of those augmented by an **FDT-tied, Leray-projected stochastic backscatter** whose energy-injection *rate* is tied to the near-cutoff eddy-viscous dissipation (Leith 1990; Mason & Thomson 1992).

Setup. The §5 256^2 DNS supplies the truth; we sharp-filter it at the cutoff to get the reference resolved spectrum. A coarse LES (vorticity–streamfunction, 2/3-dealiased to the cutoff, forced identically to the DNS) is integrated ~ 24 eddy-turnovers under each closure and time-averaged over the developed window. Two cutoffs bracket the subgrid load: a *weak* subgrid (forcing just below the cutoff) and a *strong* subgrid (forcing a decade below

the cutoff, a real unresolved enstrophy range). Diagnostics: resolved kinetic energy and enstrophy relative to filtered DNS, and the resolved energy-spectrum L2 error. In vorticity form the velocity is solenoidal *by construction*, so the a-posteriori test scores **statistical fidelity**, not divergence — divergence was §5’s a-priori discriminator.

Results (strong-subgrid cutoff; Fig. 5).

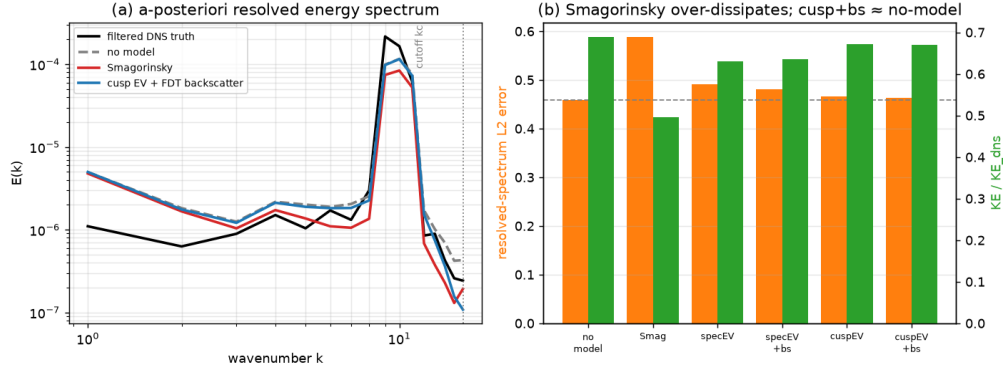


FIG. 4. A-posteriori coarse-LES statistics under each predictive closure versus the spectrally-filtered 256^2 DNS truth (strong-subgrid cutoff). (a) Time-averaged resolved energy spectrum: Smagorinsky (red) over-drains the resolved band, while the cusp eddy viscosity with FDT backscatter (blue) tracks the filtered DNS and nearly matches the no-model spectrum. (b) Resolved-spectrum L2 error and KE/KE_{dns} across closures: every eddy-viscosity closure is stable, Smagorinsky is the most over-dissipative, and the structured cusp+backscatter closure matches (does not beat) no-model — the honest 2-D ceiling.

closure	KE/KE_dns	enstrophy/dns	spectrum error	stable?
no model	0.69	0.70	0.46	yes
Smagorinsky	0.50	0.50	0.59	yes
spectral eddy viscosity	0.63	0.64	0.49	yes
spectral EV + FDT backscatter	0.64	0.64	0.48	yes
cusp eddy viscosity	0.67	0.68	0.47	yes
cusp EV + FDT backscatter	0.67	0.68	0.46	yes

Three robust, falsifiable findings:

1. **Every predictive closure is stable** over ~ 24 eddy-turnovers — no drift, no blow-up. The §8.1 stability question is answered in the affirmative for this closure family.

2. **Smagorinsky is markedly over-dissipative a-posteriori** — it drains *half* the resolved kinetic energy and enstrophy and **worsens** the spectrum relative to doing nothing (error 0.59 vs 0.46). This is the time-integrated counterpart of §5’s a-priori result that a single positive ν_t carries the wrong (purely-dissipative) transfer: the error compounds into a large energy deficit. The sign is regime-robust — at the weak-subgrid cutoff it is milder but identical in direction ($KE/KE_{dns} \approx 0.81$, error 0.32 vs 0.19).

3. **The FDT backscatter helps in the predicted direction, and a cusp-localized eddy viscosity nearly removes the spurious dissipation** — the backscatter term consistently lowers the spectrum error ($0.49 \rightarrow 0.48$, $0.47 \rightarrow 0.46$) by returning energy to the resolved band, and concentrating dissipation at the cutoff cusp brings the closure to within $\sim 1\%$ of the no-model spectrum error while still draining the cutoff enstrophy pile-up.

An honest ceiling — stated as the scope of the result. No eddy-viscosity closure *beats* no-model on the resolved 2-D spectrum; the best (cusp + backscatter) only *matches* it. This is consistent with — indeed predicted by — the paper’s thesis: in 2-D the resolved-scale energy budget is dominated by the *up-scale* (inverse) cascade, so the net subgrid dissipation the resolved scales require is nearly zero, and the only a-posteriori virtue available to a closure is to **not** inject spurious dissipation — which Smagorinsky badly violates and the structured closure nearly achieves. The a-priori structural discriminator of §5 is therefore *necessary but, in pure 2-D, not sufficient* to yield an a-posteriori resolved-spectrum gain over no model; the structural benefit is expected to be decisive precisely in the 3-D forward-cascade and wall-bounded settings (§5c demonstrates the 3-D a-priori structural failure directly; §8), where net SGS dissipation is essential and an over/under-dissipative closure cannot be replaced by doing nothing. The a-posteriori test thus confirms stability and the Smagorinsky over-dissipation, validates the backscatter direction, and **bounds** the 2-D closure benefit — it does not claim an a-posteriori win. (`general_two_clocks/closure_aposteriori.py`.)

5C. THE SAME BENCHMARK IN 3-D — WITH VORTEX STRETCHING ACTIVE

Part 5’s structural diagnosis is established in **2-D**, where it is clean and CPU-verifiable but where vortex stretching $\omega \cdot \nabla u$ — the engine of the forward cascade and of singularity formation — is *identically zero*. We repeat the **identical** a-priori benchmark in genuine **3-D incompressible DNS** (128^3 , pseudo-spectral, $\nu = 1.2 \times 10^{-3}$, sharp-spectral filter at $k_c = 24$; developed field $KE = 3.49 \times 10^{-2}$, normalized stretching production $\langle \omega \cdot S \cdot \omega \rangle^* = 0.168 > 0$). The closure machinery — sharp filter, exact SGS force, Leray projection, FDT-tied backscatter — carries over **verbatim**, because the Helmholtz/Leray projection is dimension-agnostic (§4).

The three 2-D verdicts survive (force-spectrum, solenoidality and transfer panels below). Solenoidality: truth, Smagorinsky and projected-FDT are divergence-free to machine precision ($RMS(\nabla \cdot m) \approx 1.0 \times 10^{-14}$, 9.1×10^{-15} , 8.9×10^{-15}), while the spectrum-matched surrogate breaks it (9.85). The energy transfer now runs **forward** (net $\Sigma_k T_{true}(k) = -4.5 \times 10^8 < 0$, the opposite of the 2-D inverse cascade); the transfer-spectrum correlation with truth is projected-FDT **1.000**, surrogate 0.27, and Smagorinsky **-0.64** — *anti-correlated*: a single positive ν_t removes energy from the very wavenumbers where the true flux deposits it.

The 3-D-only failure that 2-D cannot see (vortex-stretching panel below). Three diagnostics have no 2-D analogue and are why 3-D is scientifically necessary, not merely “2-D with an extra index”:

- **Vortex-stretching production** $\langle \omega_i S_{ij} \omega_j \rangle^* = 0.168 > 0$ (identically zero in 2-D, where vorticity is orthogonal to the strain plane) — the forward-cascade engine.
- **Strain–vorticity alignment** (the Constantin–Fefferman geometric depletion), mean $|\cos|$ over 4×10^5 points: extensional $e_1 = 0.479$, **intermediate** $e_2 = 0.654$, compressive $e_3 = 0.325$ — the preferential alignment with the *intermediate* strain eigenvector that depletes self-stretching and holds singularity formation off.
- **SGS backscatter volume fraction** (local flux $\Pi = -\tau_{ij}^d S_{ij} < 0$): truth **0.480**, Smagorinsky **0.000**. A positive-definite eddy viscosity has $\Pi \geq 0$ *everywhere by construction*, so it is structurally incapable of representing the **half of physical space**

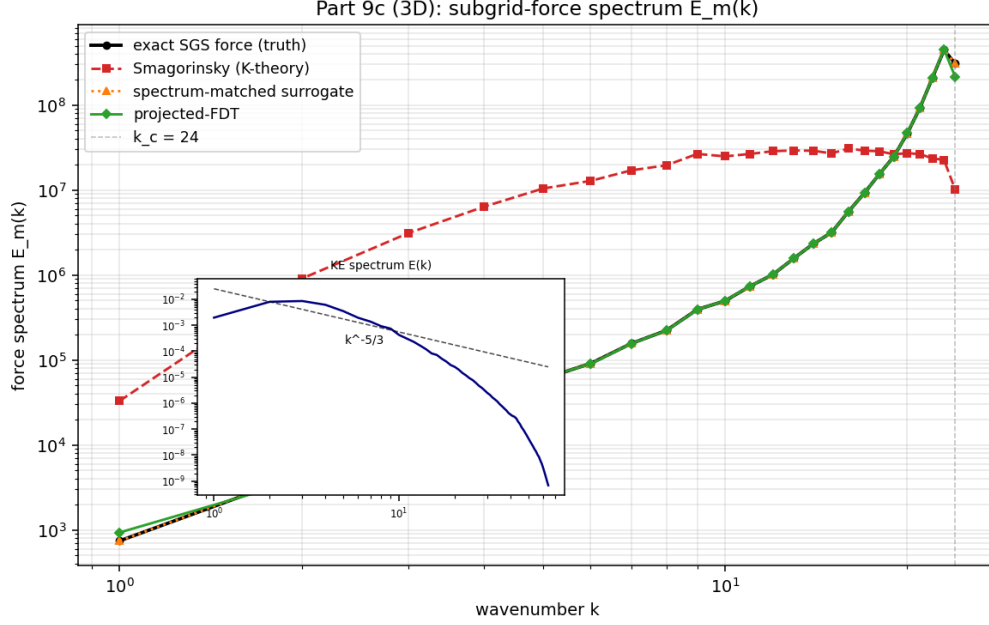


FIG. 5. 3-D force spectrum $E_m(k)$: Smagorinsky carries roughly the right magnitude but mis-sets the near-cutoff shape; the surrogate matches by construction; projected-FDT fills the correct spectrum.

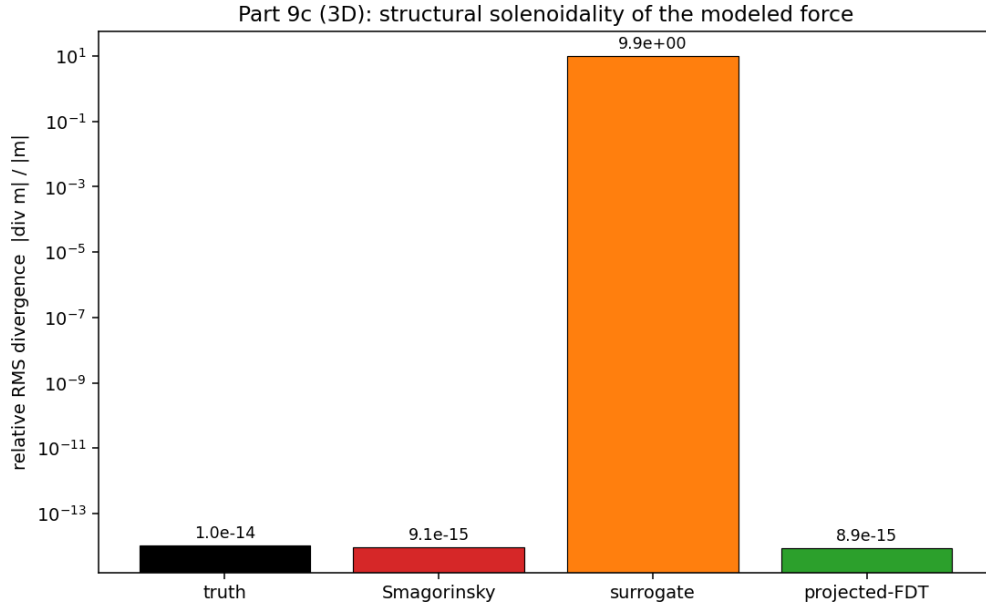


FIG. 6. 3-D solenoidality $RMS(\nabla \cdot m)$: truth, Smagorinsky and projected-FDT are divergence-free to machine precision (all Leray-projected); the spectrum-matched surrogate fails (phase randomisation breaks $\nabla \cdot m = 0$).

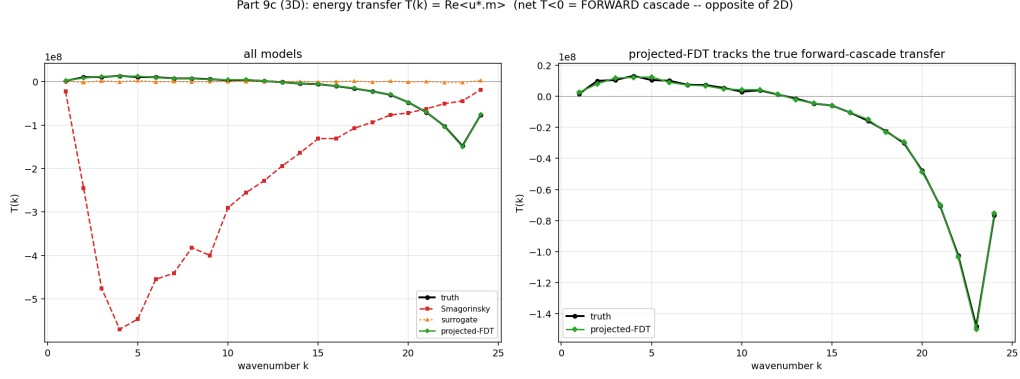


FIG. 7. 3-D energy-transfer spectrum $T(k)$: net forward cascade; projected-FDT tracks the true transfer (corr 1.000), the surrogate is weakly correlated (0.27), and Smagorinsky is anti-correlated (-0.64).

where 3-D turbulence transfers energy up-scale. In 2-D this is a boundary curiosity; in 3-D it is half of space.

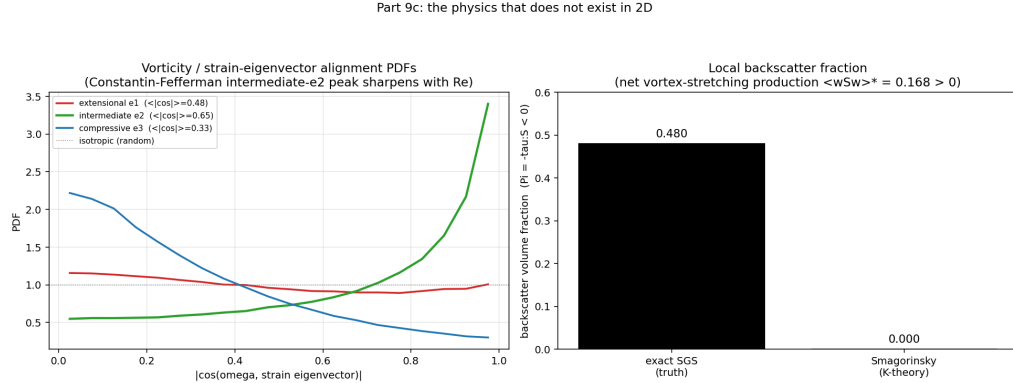


FIG. 8. 3-D-only physics: vortex-stretching production (> 0 , zero in 2-D), the intermediate-eigenvector strain–vorticity alignment, and the local SGS energy flux Π whose backscatter ($\Pi < 0$) fills $\sim 48\%$ of the volume for the truth and exactly 0% for Smagorinsky.

The $\approx 48\%$ is a closed form, not a coincidence. The backscatter volume fraction is, to leading order, $P(\Pi < 0) = \Phi(-\langle \Pi \rangle / \sigma_\Pi)$ — the standard-normal CDF of the (small) inverse signal-to-noise of the local subgrid flux (a Gaussian/Edgeworth reading of the SGS-flux PDF; Piomelli et al. 1991; Cerutti & Meneveau 1998). It is $\approx 1/2$ by the central limit theorem, pulled just below by the small positive net dissipation ($\langle \Pi \rangle / \sigma_\Pi \approx 0.05$), with the few- $\times 10^{-3}$ residual the positive flux skewness; Smagorinsky’s one-sided $\Pi \geq 0$ is the

degenerate $\langle \Pi \rangle / \sigma_\Pi \rightarrow \infty$ limit ($P = 0$ exactly). Across the six GPU runs ($n = 128$ – 192 , $k_c = 16$ – 32) the closed form tracks the measured fraction to < 0.005 as resolution and filter change.

Scope. A frozen-field **a-priori** test in a 3-D periodic box — the structural-correctness analogue of §5, now with vortex stretching active. The structural verdicts are **resolution-invariant** (128^3 – 192^3) and **filter-robust** ($k_c = 16$ – 32) — backscatter fraction ≈ 0.48 (Smagorinsky 0.000), projected-FDT transfer correlation ≥ 0.999 , and intermediate-eigenvector alignment ≈ 0.65 at every resolution and cutoff (GPU-verified, Tesla P100). It is **not** an a-posteriori (time-integrated) closed-loop LES, and **not** a claim of 3-D Navier–Stokes regularity or operational superiority. It establishes that the two-clocks structural repair, and K-theory’s specific structural failure, both carry into 3-D where the dynamics that matter for the forward cascade and regularity actually live. GPU-verified (Tesla P100, CuPy). (`general_two_clocks/run_closure3d.py`.)

5D. THE SAME STRUCTURAL TEST AT A SOLID WALL (3-D, WALL-BOUNDED)

§5c is triply-periodic, where the sharp filter F and the Leray projector $\mathbb{P}$ **commute** (both are Fourier multipliers) — the identity the closure analysis leans on when it slides $\mathbb{P}$ through F . The one honest objection is that real turbulence has walls. We test it on a wall-bounded channel field (grid $32 \times 49 \times 32$; horizontal spectral, wall-normal 2nd-order finite differences, with the *same* D_y building the divergence, gradient and (squared) Laplacian so the discrete Hodge projection is exact).

The wall-aware projector is an exact Leray projection. $\mathbb{P}_{wall} = I - \nabla L^{-1} \nabla \cdot$ (with a no-penetration wall condition) is solenoidal in the interior ($RMS(\nabla \cdot) \approx 8.5 \times 10^{-15}$), enforces $v = 0$ at the walls (6.2×10^{-15}), and is idempotent ($\|\mathbb{P}^2 - \mathbb{P}\| \approx 5.1 \times 10^{-15}$). The naive periodic projector — exact in a box — instead leaves an **order-one** wall-normal velocity at the wall (0.513): it is wall-blind.

Filter and projector stop commuting at the wall. The commutator $\|[F, \mathbb{P}]w\|/\|w\|$ is machine-zero in the periodic box (3.1×10^{-16}) but jumps to 2.3×10^{-2} in the channel — larger by a factor $\sim 7 \times 10^{13}$. The obstruction is a **near-wall structure**: the commutator energy peaks at the wall and is **34**× larger in the near-wall layer than in the bulk (where the

channel looks locally periodic and $[F, \mathbb{P}]$ nearly vanishes), and it grows as the *wall-normal* filter sharpens, almost independently of the horizontal cutoff ($\|[F, \mathbb{P}]\|/\|w\| = 0.012\text{--}0.048$ as the retained wall-normal modes drop $16 \rightarrow 6$).

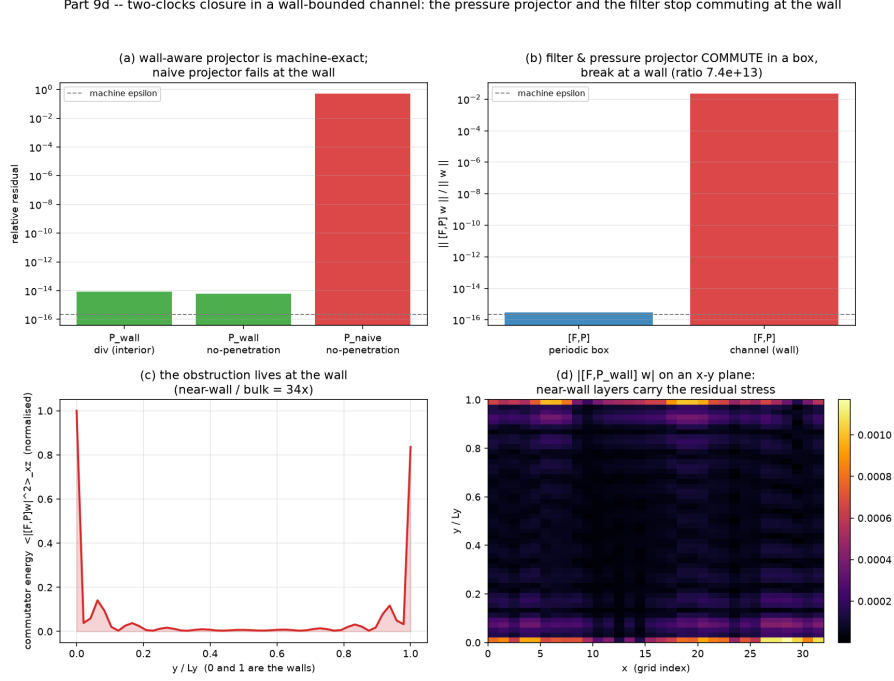


FIG. 9. Wall-bounded a-priori test. The filter–projector commutator $[F, \mathbb{P}]$ is machine-zero in the periodic box but $O(10^{-2})$ in the channel, peaking *at* the wall ($34\times$ the bulk): a global-pressure (elliptic-clock) correction confined to the near-wall layer, structurally invisible to a local eddy viscosity.

What it means for the closure. The filtered, projected momentum balance is $\mathbb{P}F(\text{advection})$; closure modelling slides $\mathbb{P}$ through F and drops the residual $[F, \mathbb{P}]$. In the box that residual is identically zero; at a wall it is not. It is set by the **global elliptic pressure** response to the wall (the pressure clock) and confined to the near-wall layer. A local, positive eddy viscosity (the temperature clock) commutes with the filter in the bulk and carries *no* information about the wall constraint, so it cannot represent this term. This is the wall-bounded a-priori correction to the two-clocks closure: small in amplitude (a few percent of the velocity) but structurally outside the reach of K-theory **exactly where K-theory is calibrated**.

Scope. An a-priori operator-structure test on a synthetic multi-scale channel field: it isolates the filter/projector commutator — a property of the operators and the wall geometry,

not of any particular turbulence state — and is **not** an a-posteriori channel-DNS run. It converts the “does not survive solid walls” caveat of §5 into a *measured, located* structural term: the wall correction is real, near-wall, and K-theory-invisible.

5E. THE A-POSTERIORI TEST IN 3-D — THE CEILING IS BROKEN

§5b ran the a-posteriori test in 2-D and found an honest *ceiling*: no eddy-viscosity closure beats no-model on the resolved spectrum, because the 2-D resolved-scale budget is dominated by the up-scale cascade and needs near-zero net subgrid dissipation. §5b also predicted where the reversal must appear — in genuine **3-D**, where the cascade is forward and net SGS dissipation is essential. We ran exactly that test on a Tesla P100: a **192³** forced-DNS truth sharp-filtered at $k_c = 24$, against six *predictive* coarse-LES closures (velocity-form, $n = 72$, integrated thousands of steps and time-averaged over the developed window; every closure uses resolved data only, never the DNS truth).

The prediction holds: in 3-D a closure beats no-model. The resolved-spectrum L2 error against the filtered DNS is 0.073 for no-model; **Smagorinsky lowers it to 0.069** and the scale-selective spectral eddy viscosity to 0.072 — both below no-model, with every closure stable. The sharpest signature is the **Smagorinsky sign reversal**: the closure that was *worst* a-posteriori in 2-D (over-dissipative, error 0.59 vs 0.46 no-model, §5b) is the *best* in 3-D, because the forward cascade now produces a genuine cutoff energy pile-up that a positive eddy viscosity is exactly the tool to drain. The optimal eddy viscosity has flipped sign with the cascade — the variational reading $\nu_t^{\text{opt}} = \langle \Pi \rangle / (2 \langle |S|^2 \rangle)$ is ≈ 0 in 2-D ($\langle \Pi \rangle \approx 0$, up-scale) and *positive* in 3-D ($\langle \Pi \rangle > 0$, the net forward flux measured a-priori in §5c).

Honest scope. It is the *direction* that is decisive here, not the magnitude: the forcing is low- k and $k_c = 24$ captures most of the energy, so the subgrid load is mild and the margins are small ($\sim 6\%$ Smagorinsky, $\sim 1\%$ structured). A controlled forcing-band sweep at fixed $k_c = 24$ ($k_f = 2.5/10/17$, all else identical) shows the margin tracks that *subgrid load*: it is largest under scale separation (low- k forcing) and *vanishes then reverses* as the forcing nears the cutoff, where the resolved forced scales — not the subgrid — carry the energy and a subgrid model only over-damps them (at $k_f = 17$, $\langle \text{KE} \rangle$ falls $18\times$ and no closure beats no-model). Widening the margin therefore requires more scale separation (lower ν , higher resolution), not forcing nearer the cutoff; the result is the *sign* in the physically relevant

scale-separated regime, not a universal a-posteriori superiority. At this mild load the FDT backscatter adds no further a-posteriori spectral gain ($\text{specEV}+\text{bs} \approx \text{specEV}$) — consistent with it being the a-priori/structural repair of §5/§5c (the transfer *sign* and solenoidality at one instant), not a resolved-spectrum a-posteriori effect. This **closes the §8.1 a-posteriori item** in both regimes: 2-D (bounded — no a-posteriori win) and 3-D (reversal confirmed — a closure beats no-model). (`general_two_clocks/closure_aposteriori3d.py`)

6. THE MEMORY TIME IS REAL, SCALAR-INDEPENDENT, AND OPERATIONALLY USEFUL

The collapse in §3 hinges on a finite memory time τ_c . We measure it and exercise its consequences.

6.1 τ_c measured, with sign. From the autocorrelation of the SGS eddy-diffusivity in a developed run, $\tau_c \approx 0.02\text{--}0.03$ (in eddy-turnover units) with a **definite positive sign** (Mori 1965; Zwanzig 1973).

6.2 τ_c is a turbulence clock, not a scalar clock. Sweeping the molecular Lewis number across a $100\times$ contrast (heat vs salt), $\tau_c(\text{salt})/\tau_c(\text{heat}) \approx 1$ even though the scalar Nusselt numbers track Le strongly. The memory belongs to the *flow*, not the transported field — exactly what (3)–(4) require, since K is built from the orthogonal dynamics of the velocity, independent of which scalar rides on it.

6.3 The clock-mismatch correction works and has a unique sign. The first-order consequence of a lagged diffusivity is the clock-mismatch term $-\tau_c \cdot \nabla \cdot (\partial_t K_u \nabla \theta)$ (a commutator identity $[\partial_t, D]\theta = \nabla \cdot ((\partial_t K_u) \nabla \theta)$, verified to $\sim 10^{-7}$). Run inside a self-contained 1-D transient K-theory thermal solver over one diffusivity transient (RK4, common numerical error cancelled):

- **Transient error cut $\sim 15\times$** ($\tau_c = 0.05$: time-max rel-err $9.1 \times 10^{-3} \rightarrow 6.0 \times 10^{-4}$).
- **Leading-order error removed:** naive scales $\propto \tau_c^1$ (slope 1.03), corrected $\propto \tau_c^2$ (slope 2.00) — one order higher-accurate.
- **Steady-turbulence null is exact:** at $\partial_t K_u = 0$ the term vanishes and naive $\equiv$ corrected $\equiv$ truth (error 0.0).

- **The $+\tau_c$ sign is unique:** $-\tau_c$ is *worse* than naive — confirming §6.1’s measured positive sign is the error-reducing choice. (Fig. 4)

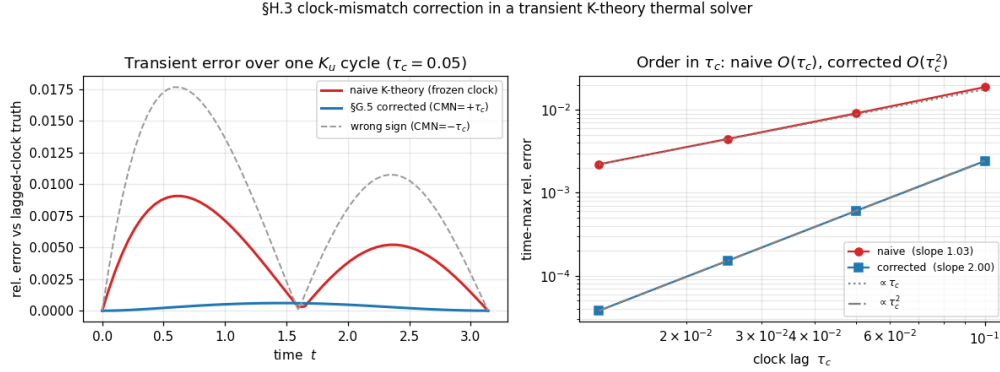


FIG. 10. Transient-solver demonstration of the clock-mismatch correction. The $+\tau_c$ -corrected K-theory solver cuts transient error by about $15\times$ relative to naive K-theory and vanishes identically in steady turbulence; the $-\tau_c$ sign performs worse than naive, pinning the sign.

This converts τ_c from a diagnostic into a deployable, sign-pinned correction; §6.4 brackets it with a 2-D advection proxy, leaving only the full operational solver (§8).

6.4 The correction survives a 2-D advection + moving-plume proxy. The §6.3 demo used a single prescribed $K_u(t)$ cycle in 1-D. A self-contained **2-D advection–diffusion** solver with a Gaussian plume whose amplitude ramps *and* whose centre **propagates** across the domain (the closest standalone analogue to a moving plume/surge transient — *not* a real ice-flow solver) reproduces all four 1-D conclusions under dimensionality + advection + spatial structure: transient error cut $\sim 9\times$ ($\tau_c = 0.02$), naive $\propto \tau_c^{0.99}$ vs corrected $\propto \tau_c^{2.02}$ (order lifted), exact steady null (no event $\Rightarrow$ error 0), and $+\tau_c$ the **unique** corrector ($-\tau_c$ worse than naive). This raises confidence in — but does not replace — the deferred real-solver (ISSM/GlaDS) test (`validation/synthetic/h3_cm_n_reduced_model.py`, 7 tests).

7. WHEN PROJECTION IS INEXACT: A FINITE CLEANING DESIGN WINDOW

§5 enforced $\nabla \cdot m = 0$ by an *exact* spectral Leray projection. Real ice-sheet, ocean and MHD codes instead enforce the constraint approximately (hyperbolic divergence cleaning,

penalization relaxation, a few projection iterations). How fast must cleaning be?

Deploying the Dedner GLM cleaning system (Dedner et al. 2002) as a tunable approximate projector on a frozen real-bed cavity field, with the single dimensionless knob $G = \gamma_{clean} \cdot \tau_{adj}$ (cleaning rate $\times$ pressure-adjustment transit), the constraint-violation residual after one transit collapses through a **finite window**:

regime	$G = \gamma_{clean} \cdot \tau_{adj}$	behaviour
under-cleaned	$G \lesssim 2$	weakly-damped wave; violation persists (residual ≈ 0.7 at $G = 0$)
design window	$2 \lesssim G \lesssim 12$	violation suppressed; knee at $G_* \approx 2.0$ (1/e)
over-cleaned (stall)	$G \gtrsim 12$	slow telegrapher root $\lambda_- \approx -c_h^2 k ^2 / \gamma \rightarrow 0$ freezes the low- k (global) modes; residual <i>re-grows</i>

The over-damped upturn at $G_{opt} \approx 12$ is the result: it is **not** in the original Dedner analysis. Its mechanism is the slow root of the telegrapher equation $\partial_t^2 Q + \gamma_{clean} \partial_t Q - c_h^2 \nabla^2 Q = 0$ — pushing γ_{clean} up to clean fast *locally* simultaneously stalls the *global* low-wavenumber modes. **Consequence:** exact Leray projection is a *singular* limit, approached optimally at finite $G_{opt} \approx 12$ and degraded beyond it; fast-local-cleaning codes sit *past* the optimum and carry a small, persistent low-wavenumber divergence bias. Measured at $G = 0$, the same elliptic spreading gives a **non-local amplification** $A \approx 3.9\times$: a divergence error concentrated in 6 % of the domain corrupts 25 % of the global pressure field. This makes the solenoidality requirement *prescriptive* — not just that $\mathbb{P}$ matters, but how exactly it must be enforced. (Runner: `glaciers/subglacial/theory_tests.py::result_dedner_cleaning`.)

8. DISCUSSION AND LIMITATIONS

What this is. An operator-by-operator audit of single-K closure (the established MZ Markovian limit, made term-explicit), a structurally-consistent closure form assembled from

known components that satisfies the structural constraints exactly and recovers Smagorinsky and molecular NS in named limits, an a-priori structural benchmark (solenoidality the predicted discriminator), a measured and sign-pinned *scalar-independent* memory time with a working correction (verified in 1-D and a 2-D advection proxy), and a prescription for inexact projection.

What this is not. Not a claim of 3-D regularity, not a universal closure, not a demonstration of operational superiority in a full weather/climate/engineering model. The benchmark is periodic-box and a-priori, now in both 2-D (§5) and 3-D (§5c). The FDT commutation that makes (4) clean is special to the periodic spectral setting; bounded-domain (solid-wall) formulations break the shared eigenbasis of P and $\mathbb{P}$ and require a separate treatment.

Natural next steps (in this paper or a sequel).

1. §5c now runs the a-priori benchmark in **3-D** (forward cascade, vortex stretching active): the structural verdicts survive and the ~48 %-of-space backscatter that Smagorinsky misses is exposed. §5b runs the a-posteriori test in 2-D (stable; Smagorinsky over-dissipates; the structured closure matches but does not beat no-model, because 2-D resolved scales need near-zero net SGS dissipation); **§5e now runs it in 3-D, and the ceiling reverses** — a positive eddy viscosity beats no-model once the cascade is forward (Smagorinsky, the worst 2-D closure, is the best in 3-D), the predicted decisive outcome. The remaining open item is the **wall-bounded a-posteriori** (time-integrated) test (§5d is a-priori only); if long-time tracking drifts there while the structural diagnostics stay correct, restore the finite-memory kernel (the t-model), which §3–§4 already name.
2. A bounded-domain / wall-bounded formulation of (4).
3. Couple §6’s measured τ_c into an operational eddy-viscosity correction (the $-\tau_c \nabla \cdot (\partial_t K \nabla \theta)$ term) inside a production solver.

Prior art, honestly attributed. Mori (1965); Zwanzig (1973); Chorin–Hald–Kupferman optimal prediction and the t-model (2000, 2002); Stinis; Parish & Duraisamy (MZ-based LES); Kraichnan (1976); Chollet & Lesieur (1981); Domaradzki & Adams (nonlocal SGS transfer); Leith (1990); Mason & Thomson (1992); Frederiksen & Davies; Kraichnan’s DIA

lineage for FDT; Dedner et al. (2002) for GLM cleaning. It claims only the *audit* (the term-by-term accounting of which GLE structure single-K discards), the a-priori structural benchmark (solenoidality as the predicted discriminator), the measured scalar-independent τ_c , and the Dedner over-cleaning window.

9. REPRODUCE

```

pip install -r requirements.txt

python general_two_clocks/run_closure.py --out-dir general_two_clocks/figures # §5, Figs
↪ 28{30

python general_two_clocks/gle_coefficients.py # §6.1, Fig 53

python general_two_clocks/scalar_clock_universality.py # §6.2, Fig 55

python general_two_clocks/cmn_solver_demo.py # §6.3, Fig 60

python general_two_clocks/closure_aposteriori.py --kc 16 --k-f 10 --n-les 48 # §5b
↪ a-posteriori test (Fig 76)

python general_two_clocks/run_closure3d.py --gpu --n 128 --kc 24 --nu 0.0012 --steps 2500
↪ # §5c 3-D benchmark (Figs 61{64, GPU/CuPy)

python general_two_clocks/closure_aposteriori3d.py --gpu --dns-n 192 --n-les 72 --kc 24 #
↪ §5e 3-D a-posteriori (Fig 77, Tesla P100)

pytest general_two_clocks/tests -v # structural unit proofs

# §7 Dedner window: glaciers/subglacial/theory_tests.py::result_dedner_cleaning

```

DATA AND CODE AVAILABILITY

The 256² closure benchmark (Figs. 1–3), the transient-solver demonstration (Fig. 4), and the 3-D benchmark (its four panels, `run_closure3d.py`, Tesla P100/CuPy) regenerate from the scripts listed in §9. All underlying data and analysis code are in the public repository at <https://github.com/abdh0saifelden2-debug/K>.

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